

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Trigonometrical Equations

Q1

If α and β are the roots of equation $(k+1) \tan^2 x - \sqrt{2} \lambda \tan x = 1 - k$ and $\tan^2 (\alpha + \beta) = 50$. Find value of λ .

यदि α तथा β समीकरण $(k+1) \tan^2 x - \sqrt{2} \lambda \tan x = 1 - k$ के मूल हैं तथा $\tan^2 (\alpha + \beta) = 50$ तो λ का मान है—

- (1) 10 (2) 5 (3) 7 (4) 12

7th Jan Morning

Sol

(1)

$$(k+1) \tan^2 x - \sqrt{2} \lambda \tan x + (k-1) = 0$$

$$\tan \alpha + \tan \beta = \frac{\sqrt{2} \lambda}{k+1}$$

$$\tan \alpha \tan \beta = \frac{k-1}{k+1}$$

$$\tan (\alpha + \beta) = \frac{\frac{\sqrt{2} \lambda}{k+1}}{1 - \frac{k-1}{k+1}} = \frac{\sqrt{2} \lambda}{2} = \frac{\lambda}{\sqrt{2}}$$

$$\tan^2 (\alpha + \beta) = \frac{\lambda^2}{2} = 50$$

$$\lambda = 10$$

02

Let θ_1 and θ_2 (where $\theta_1 < \theta_2$) are two solutions of $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$, $\theta \in [0, 2\pi)$ then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is

equal to

माना θ_1 और θ_2 (जहाँ $\theta_1 < \theta_2$) समीकरण $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$, $\theta \in [0, 2\pi)$ के दो हल हैं तब $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ का

मान है—

- (1) $\frac{\pi}{3}$ (2) $\frac{2\pi}{3}$ (3) $\frac{\pi}{9}$ (4) $\frac{\pi}{3} + \frac{1}{6}$

Sol

(1)

$$2\cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$$

$$\frac{2\cos^2 \theta}{\sin^2 \theta} - \frac{5}{\sin \theta} + 4 = 0$$

$$2\cos^2 \theta - 5\sin \theta + 4\sin^2 \theta = 0, \sin \theta \neq 0$$

$$2\sin^2 \theta - 5\sin \theta + 2 = 0$$

$$(2\sin \theta - 1)(\sin \theta - 2) = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \int_{\pi/6}^{5\pi/6} \cos^2 3\theta \, d\theta = \int_{\pi/6}^{5\pi/6} \frac{1 + \cos 6\theta}{2} \, d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin 6\theta}{6} \right]_{\pi/6}^{5\pi/6} = \frac{1}{2} \left[\frac{5\pi}{6} - \frac{\pi}{6} + \frac{1}{6}(0 - 0) \right] = \frac{1}{2} \cdot \frac{4\pi}{6} = \frac{\pi}{3}$$

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